

App.jsx Explanation for Interview

Overview

``src/App.jsx`` is the root React component for the Khoje Khatam web app. It sets up app state, filtering logic, user login, and renders the main page structure.

Imports

- ``useMemo``, ``useState``, ``useEffect`` from React for state, memoized values, and lifecycle effects.
- ``useLocation`` from ``react-router-dom`` to read query string parameters.
- ``contentItems`` from ``./data/pyqs.js`` is the dataset used for search and filtering.
- Components imported:
 - ``SearchBar`` for search and filters.
 - ``ContentList`` for list of filtered study items.
 - ``ContentView`` for displaying the selected item.
 - ``ProfileMenu`` for login / user account UI.
 - ``TermsAndConditions`` and ``PrivacyPolicy`` for modal dialogs.
 - ``BranchFilter`` for branch selection.

State Variables

- ``search``: text search input.
- ``selectedId``: id of the currently selected content item.
- ``filterTopic``: current topic filter.
- ``filterSubject``: current subject filter.
- ``filterSemester``: current semester filter.
- ``filterYear``: current year filter.
- ``filterBranch``: current branch filter.
- ``isLoggedIn``: whether the user is logged in.
- ``user``: logged-in user profile data.
- ``showProfile``: whether the profile modal is visible.
- ``showTerms``: whether terms dialog is visible.
- ``showPrivacy``: whether privacy dialog is visible.
- ``isHeaderMinimized``: whether header shrinks on scroll.

Memoized Lists

- ``branches``: unique branch names from ``contentItems``, with ``All`` first.
- ``topics``: unique topic names from ``contentItems``, with ``All`` first.
- ``years``: unique year values from ``contentItems``, with ``All`` first.

These values are memoized so they don't recompute on every render unless the dependencies change.

Filtering Logic

- ``filteredItems`` is computed from ``contentItems``.
- For each item, the code checks:
 - ``matchesSearch``: whether search text appears in title, subject, content, or

notes.

- ``matchesSubject``: whether the selected subject filter matches.
- ``matchesSemester``: whether the selected semester filter matches.
- ``matchesTopic``: whether the selected topic filter matches.
- ``matchesYear``: whether the selected year filter matches.
- ``matchesBranch``: whether the selected branch filter matches.
- The item is included only if all checks pass.

Selected Item

- ``selectedItem`` is the item whose ``id`` matches ``selectedId``.
- If no selected item exists, it falls back to the first item from ``filteredItems``.
- If there is still no item, it becomes ``null``.

Login Handlers

- ``handleLogin(userData)``: stores the user data and marks ``isLoggedIn`` true.
- ``handleLogout()``: clears the user data and closes the profile menu.

Header Scroll Effect

- ``useEffect`` adds a scroll listener.
- If the window scroll position is above 100px, ``isHeaderMinimized`` is false.
- If it is below 100px, ``isHeaderMinimized`` becomes true.
- This enables a smaller header while scrolling.

Query Parameter Handling

- The app reads URL query parameters using ``useLocation()``.
- It supports:
 - ``branch`` to preselect a branch.
 - ``subject`` to preselect a subject.
 - ``semester`` to preselect a semester.
- When query params exist, the app applies them to filters and selects the matching item.

Rendered Layout

The component returns the app structure:

Header

- Displays the app logo and title.
- Contains ``ProfileMenu`` with login/account controls.

About Section

- Contains marketing text describing the app.
- Shows feature highlights and an "Explore Resources" button.

Branch Filter

- Renders ``BranchFilter`` to choose the branch.
- ``branches`` is passed from the memoized list.

- Selected branch changes update `filterBranch`.

Main Content

- `SearchBar` shows search input and filter controls.
- `ContentList` shows the list of filtered study content.
- `ContentView` shows the details of the selected item.
- If no item is selected, it shows an empty state message.

Footer

- Includes brand text and quick links.
- Links open the Terms and Privacy dialogs.
- Social icons are included.
- Copyright text is shown.

Modals

- `TermsAndConditions` is rendered when `showTerms` is true.
- `PrivacyPolicy` is rendered when `showPrivacy` is true.

Key Interview Points

- This file is responsible for app-level state and page layout.
- It uses React hooks for state, memoization, and side effects.
- The filtering logic is centralized in `filteredItems`.
- The component passes state and handlers down to child components.
- It supports deep linking through URL query parameters.
- It has a responsive header effect based on scrolling.

Summary

`App.jsx` acts as the root container for the Khoje Khatam application, coordinating:

- search filtering,
- branch/topic/year/semester filters,
- item selection and display,
- login/profile state,
- modal visibility,
- and page rendering.

This explanation is suitable for preparing interview notes or converting into a PDF document.